Arsen Khangueldyan

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Process-automation engineer with 7 years building and operating production systems in highly regulated environments (defense, automotive, enterprise). Microsoft Azure DevOps Engineer Expert (AZ-400). Hands-on with AI-enabled workflow automation, data integration, requirements engineering, and SDLC / V&V discipline. Recent focus: AI agents that triage governance workflows (ATO, NTAP, compliance review).

SELECTED PROJECTS

ATO Copilot — AI Triage for ATO/NTAP Governance Workflows 2026

github.com/smartpneucontact-sketch/ATO_Copilot

- End-to-end AI workflow tool: free-text NTAP request goes in, structured ATO package comes out — Approved Technology List check, NIST 800-53 control mapping, reference-architecture review, risk classification, cycle-time estimate, recommended decision (ATO-APPROVED / CONDITIONS / DENIED / ARB-ROUTING).
- Agentic AI loop on Claude Sonnet 4.6 with RAG over a synthetic governance corpus (ATL entries, NIST control families, reference architecture patterns, prior ATO precedents). Three custom tools: retrieve, classify_risk, estimate_cycle_time.
- Production-target architecture explicitly mapped onto State Street's stack: Microsoft Copilot Studio + Azure OpenAI, ServiceNow workflow + Power Automate connector, Azure AI Search / SharePoint, Flexera ITAM data, Splunk + Azure Monitor telemetry.
- FastAPI service with visitor-notify (Resend HTTPS API), modern dark UI with semantic rendering of every output field, Railway deploy ready.

PROFESSIONAL EXPERIENCE

Hyperion Nov 2022 – Jun 2025

Head of Engineering · Yerevan, Armenia

- Shipped a fully autonomous AI defense drone — acquired by strategic buyer; owned the full engineering lifecycle across AI software, computer vision, firmware, electronics, mechanical, and composites.
- Operated under strict regulated-industry process discipline — requirements engineering, design reviews, traceability matrix, risk register, formal V&V, ITIL-shaped change management — directly applicable to State Street's SDLC + ATO / ARB governance work.
- Led a multidisciplinary team and managed cross-functional dependencies with vendors, regulators, and customer stakeholders.
- Built edge-deployed computer vision for GPS-denied navigation at 96% accuracy; validated across 3,000 flight-test hours.

Deloitte Apr 2021 – Jun 2022

Data Analyst · Luxembourg

- Led migration of multi-source enterprise data into a unified Azure-hosted repository with static + dynamic metadata governance; exposed via REST API to a single-page web app. Standard governance / data-architecture engagement for a regulated client.
- Evaluated nine AutoML platforms (IaaS / PaaS / SaaS) against predefined test protocols; produced selection rationale, risk-control mapping, and executive presentations.

Forschungsgesellschaft Umformtechnik mbH May 2019 – Mar 2021

Data Engineer · Stuttgart, Germany

- Shipped a tool-wear classifier (CNN, 82% accuracy) into TRUMPF Group's next-generation punching machines; co-authored a peer-reviewed publication on CNN-based tool-wear classification (Feb 2020). Production deployment with the full QA and approval cycle.
- Built a sensor-fusion IoT pipeline (force, distance, sound, lubrication) and an ML failure-prediction algorithm on Audi AG data — connecting operational telemetry into an actionable signals layer, the data-architecture pattern State Street's JD asks for.

Audi AG Apr 2018 – Dec 2018

Software Developer · Neckarsulm, Germany

- Built and deployed an Oracle APEX low-code web application that digitalized die-cast tooling improvements; rolled out internationally across Audi sites. Process-automation / paper-to-digital pattern.

TECHNICAL SKILLS

AI Coding Assistants
Claude Code (deep proficiency — shipping production agentic systems with it), GitHub Copilot, Cursor, Devin AI awareness; AI-pair-programming workflows, prompt-driven development, structured tool-use design

LLMs & Agentic AI
Anthropic Claude (Opus / Sonnet / Haiku), OpenAI GPT-4o family, Azure OpenAI, AWS Bedrock; Agentic RAG, multi-tool agent loops, ReAct pattern, structured outputs (JSON Schema), function / tool calling, multi-agent orchestration, prompt engineering

RAG, Vector Stores & Embeddings
Hybrid retrieval (BM25 + dense, Reciprocal Rank Fusion), chunking strategies (semantic, heading-based, sliding-window), citation grounding, retrieval eval (context precision, faithfulness, answer relevancy); Chroma, OpenSearch kNN, Azure AI Search, Pinecone awareness; OpenAI / Voyage / Cohere embeddings

LLMOps
JSONL tracing (run_id / span / token / cost), eval-gated CI with YAML test cases + scorer functions, drift detection (rolling distribution monitors on response length / citation count / escalation rate), prompt versioning, prompt-injection red-teaming, cost & latency monitoring per agent run

Microsoft Copilot stack
Microsoft Copilot Studio architecture, GitHub Copilot Enterprise, Microsoft 365 Copilot, Power Platform AI Builder, Power Apps + Power Automate (applied via prototype), Dataverse-shaped integration

Microsoft / Azure
Azure DevOps Engineer Expert (AZ-400), Azure Administrator (AZ-204), Azure Fundamentals (AZ-900), Azure Data Factory, Active Directory, custom connector design

Governance & Risk
Reference architecture review, NIST 800-53 control mapping, audit-readiness artifacts, security risk processes, ITIL processes, regulated-industry V&V discipline (defense, automotive)

Engineering & Data
Python (expert), FastAPI, C/C++, Docker, Kubernetes, Terraform, GitHub Actions, AWS, SQL, ETL / ELT design, REST APIs, sensor-fusion pipelines

Classical ML & Computer Vision
PyTorch, HuggingFace Transformers, YOLO, ResNet-style CNNs, CUDA, edge inference, model quantization (autonomous-system production deployment)

CERTIFICATIONS

Microsoft Azure DevOps Engineer Expert (AZ-400) · Microsoft Azure Administrator (AZ-204) · Microsoft Azure Fundamentals (AZ-900) · AWS Cloud Practitioner · TensorFlow Developer · Deep Learning in Computer Vision

EDUCATION

Ecole Centrale d'Electronique — Paris, France
M.S. Computer Science & Engineering — Jun 2019
B.S. Mathematics & Electronics — Jun 2017

MIT Professional Education — Dec 2025 – Sep 2026
Digital Transformation in the AI Age (in progress)

LANGUAGES

English (proficient) · French (fluent) · German (fluent) · Armenian (native)